

Symbolic Logic and Truth Tables

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Determine if a sentence is a statement.
- ☐ Translate verbal statements into symbolic form using logical notation and vice versa.
- ☐ Determine the truth value of a statement.
- ☐ Construct a truth table for a statement.

Motivation

Consider the following situation:

Holmes announces, “Watson, I have uncovered the following facts:

- If Mrs Smith is lying then Moriarty has not escaped.
- Either Moriarty is dead or he is really Jones.
- If Moriarty is really Jones then he has escaped.
- Mrs Smith is lying.

“Good Lord Holmes,” replied Dr Watson, “what can you make of all this?”

“Elementary my dear Watson, Moriarty is dead.”

Is Holmes correct?

Statements

A **statement** (or **proposition**) is any sentence that is unambiguously true or false.

Example: Which of the following is a statement?

- p : $1 + 1 = 2$. A true statement.
- q : 2 is an even number. A true statement.
- $2 > 5$. A false statement.
- The number x is positive. Not a statement.
- Kuala Lumpur is the capital of Malaysia. A true statement.
- The sun rises in the West. A false statement.
- Show that x is an integer. Not a statement.
- Is the integer $(2^{10} - 1)$ even? Not a statement.
- Sunday is the best day of the week. Not a statement.

Typically, a statement is denoted by a lowercase letter, for example p or q , and these letters are called **statement variables**.

Compound statements

A **compound statement** is a sentence made up of two or more statements.

We decide if a compound statement is true or false depending on whether its component statements are true or false and the type of connective we used.

For example, ^{P:} “1 is an odd integer and 2 is an even integer” is a compound statement with the component statements “1 is an odd integer” and “2 is an even integer”.
 (Note: In the original image, 'and' is boxed and labeled 'connective' with an arrow.)

We usually use an uppercase letter to represent a compound statement, for example P or Q .

eg. of connective :

- NOT
- AND
- OR
- IF - THEN
- IF AND ONLY IF

Truth table

A **truth table** is a breakdown of all the possible truth values returned by a logical expression. A truth value is typically either **true (T)** or **false (F)**.

When considering compound statements, we use truth tables to exhaust all possible combinations of the component simple statements.

Example:

Power is on	Electricity is present	Air conditioner is working
T	T	T
T	F	F
F	T	F
F	F	F

There are 2^n rows in the truth table when considering the compound statement consists of n component statements.

(NOT INCLUDING THE TITLE ROW)

Truth table

$n=1$ (p)

P
T
F

2^1 rows

$n=2$ (p, q)

P	q	$p \wedge q$
T	T	
T	F	
F	T	
F	F	

$2^2 = 4$ rows

$n=3$ (p, q, r) $2^3 = 8$ rows

	P	q	r	$(p \wedge q) \vee r$
0	T	T	T	
1	T	T	F	
2	T	F	T	
3	T	F	F	
4	F	T	T	
5	F	T	F	
6	F	F	T	
7	F	F	F	

$n=4$ (p, q, r, s) $2^4 = 16$ rows.

P	q	r	s
T	T	T	T
T	T	T	F
T	T	F	T
T	T	F	F
T	F	T	T
T	F	T	F
T	F	F	T
T	F	F	F
F	T	T	T
F	T	T	F
F	T	F	T
F	T	F	F
F	F	T	T
F	F	T	F
F	F	F	T
F	F	F	F

T	F	T	T
T	F	<u>T</u>	F
T	F	F	T
T	F	F	F
<u>T</u>	<u>F</u>	<u>F</u>	<u>F</u>
F	T	T	T
F	T	<u>T</u>	F
F	T	F	T
F	T	F	F
<u>F</u>	<u>F</u>	<u>T</u>	<u>T</u>
F	F	T	F
F	F	<u>T</u>	<u>F</u>
F	F	F	T
F	F	F	F

Negation “not”

The **negation** of a statement is a statement that is true precisely when the original statement is false.

The negation of the statement p denoted by $\sim p$, and can be read as “not p ”. (Some references might use the notation $\neg p$.)

The truth table of $\sim p$ is as follows:

p	$\sim p$
T	F
F	T

Negation “not”

Example:

p	$\sim p$
$3 + 3 = 6.$ T	$3 + 3 \neq 6$ F
13 is a prime number. T	13 is not a prime number. F
All integers ending in 3 are prime.	Not all integers ending in 3 are prime.

Some integers ending in 3 are not prime

It is false that all integers ending in 3 are prime.

Conjunction “and”

The **conjunction** of two statements is a compound statement that is true precisely when both the component statements are true.

The conjunction of the statements p and q is denoted by $p \wedge q$, and can be read as “ p and q ”.

The truth table of $p \wedge q$ is as follows:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: (Kuala Lumpur is in Malaysia) and (zero is positive.) F

T F

Some special occurrences (But and Neither-Nor)

Sometimes, the word “but” can be used in place of “and”.

e.g. The integer 2 is prime but it is even.

equiv. to “(The integer 2 is prime) and (2 is even).” True
 $\underbrace{\hspace{1cm}}_T \quad \underbrace{\hspace{1cm}}_T$

The phrase “neither p nor q ” means “not p and not q ” ($\sim p \wedge \sim q$).

Note: $\sim p \wedge \sim q$ DOES NOT mean $\sim (p \wedge q)$ and $\sim p \wedge q$ DOES NOT mean $\sim (p \wedge q)$.
||| $\sim p \vee \sim q$ * Bracket is important !!

e.g. Neither π nor $\sqrt{2}$ is integer.

equiv. to “ π is not integer and $\sqrt{2}$ is not integer”

p : π is an integer

q : $\sqrt{2}$ is an integer

$\sim p \wedge \sim q$

p : π is not integer

q : $\sqrt{2}$ is not integer

$p \wedge q$

Disjunction “or”

The **disjunction** of two statements is a compound statement that is true precisely when either (or both) of the two component statements are true.

The disjunction of the statements p and q is denoted by $p \vee q$, and can be read as “ p or q ”.

The truth table of $p \vee q$ is as follows:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Disjunction “or”

Example:

- ① 2 or -2 is zero. equiv. $(\overset{F}{2 \text{ is zero}}) \text{ or } (\overset{F}{-2 \text{ is zero}})$. **False.**
- ② $(\overset{0}{\sin \pi})$ or $(\overset{1}{\cos \pi})$ is zero. **True.**
- ③ $2 \leq 2$. eq: $(\overset{F}{2 < 2}) \text{ or } (\overset{T}{2 = 2})$ **True.**

Remark:

The notation for inequalities involves “and” and “or” statements:

Let a, b, c be some integers

$a \leq b$ means $a < b$ **or** $a = b$.

$a < b < c$ means $a < b$ **and** $b < c$.

In mathematics, the word “or” always mean the **inclusive or** (p or q or both). To express **exclusive or**, we use p or q **but not both**.

Conditional Statement

A **conditional statement** (or **implication statement**) has the form “If p is true, then q is true”.

The conditional statement “If p is true, then q is true” is denoted by $p \rightarrow q$, and can be read as “ p implies q ”.

The truth table of $p \rightarrow q$ is as follows:

p only if q
 $p \rightarrow q$
If q does not happens
then p does not happens

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Eg: If (^Ptoday is Thursday) then
(I will be available in my office.)^q

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

For $p \rightarrow q$, p is called the **hypothesis** (or **antecedent**) and q is called the **conclusion** (or **consequent**).

Conditional Statement

All of the following are equivalent: $(p \rightarrow q)$

- If p is true, then q is true (If p , then q)
- p implies q
- Whenever p is true, q is true (q whenever p)
- ✓ • q is true if p is true (q if p)
- ✓ • p is true only if q is true (p only if q)
- q follows from p
- ✓ • q unless $\sim p$
- p is a sufficient condition for q
- q is a necessary condition for p
- A sufficient condition for q is p
- A necessary condition for p is q

If a number is divisible by 4, then it is an even number.

⇒ Whenever a number is divisible by 4, it is an even number.

⇒ If not q , then not p .

A number is divisible by 4 only if it is an even number.

✓ • q unless $\sim p$ ⇒ A number is even unless it is not divisible by 4.

Conditional Statement

Example:

p only if q

- ① (The moon is square)^{p F} only if ^q (the sun rises in the East.) *True.*
- ② (1 and 3 are prime)^{q P} if ^{P T} (1 multiply 3 is prime.) *p → q False.*
- ③ ((sin π)(cos π) = 0)^{q T} when ^{P T} (sin π = 0 or cos π = 0.) *True.*
- ④ (February has 30 days)^{P F} implies ^q (March has 31 days.) *True.*

Remark:

If the hypothesis p is false, $p \rightarrow q$ say nothing about what will happen. In this case, we consider $p \rightarrow q$ is true, regardless of its conclusion, and this is called true by default.

A true conditional statement DOES NOT mean a conditional with true conclusion.

If 2 or T_2 is negative, then $(T_2 - 2)$ is negative. True

If 2 or T_2 is negative, then $(T_2 + 2)$ is negative. True statement.

Conditional Statement

The **negation** of $p \rightarrow q$ is $p \wedge \sim q$. $[\sim(p \rightarrow q) \equiv p \wedge \sim q]$

The **converse** of $p \rightarrow q$ is $q \rightarrow p$.

The **inverse** of $p \rightarrow q$ is $\sim p \rightarrow \sim q$.

The **contrapositive** of $p \rightarrow q$ is $\sim q \rightarrow \sim p \equiv p \rightarrow q$

Example: Write down the negation, converse, contrapositive and inverse of the following conditional statement.

If ^p(2 and 3 are prime numbers) ^qthen (2 + 3 is prime)

Negation: 2 and 3 are prime numbers and ^(but) 2+3 is not prime.

Converse: If 2+3 is prime, then 2 and 3 are prime.

Inverse: If 2 or 3 is not prime, then 2+3 is not prime.

Contrapositive: If 2+3 is not prime, then 2 or 3 is not a prime number.

If ^p($x^2 = 2$) ^qthen ($x = \pm\sqrt{2}$)

Negation: $x^2 = 2$ and $x \neq \pm\sqrt{2}$.

Converse: If $x = \pm\sqrt{2}$, then $x^2 = 2$.

Inverse: If $x^2 \neq 2$, then $x \neq \pm\sqrt{2}$.

Contrapositive: If $x \neq \pm\sqrt{2}$, then $x^2 \neq 2$.

Biconditional Statement

A **biconditional statement** has the form “ p is true if and only if q is true”. It means both “If p is true, then q is true” and “If q is true, then p is true”.

The biconditional statement “ p is true if and only if q is true” is denoted by $p \leftrightarrow q$, and can be read as “ p if and only if q ”.

The truth table of $p \leftrightarrow q$ is as follows:

$p \rightarrow q$	$q \rightarrow p$	p	q	$p \leftrightarrow q$
T	T	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	F	F	T

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$\equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

T
F
F
T

Biconditional

All of the following are equivalent:

- $p \leftrightarrow q$
- p is true if and only if q is true *p if and only if q*
- p iff q (The term “iff” is often used as shorthand for “if and only if.”)
- p is (logically) equivalent to q
- p implies q and q implies p
- p is a necessary and sufficient condition for q

Example:

- ① (2 is prime) ^T if and only if (it is multiple of 2.) ^T *True.*
- ② ($\sqrt{2}$ is negative) _F if and only if (2 is negative.) _F *True.*
- ③ (2 is an even number) _T iff (3 is an even number) _F *False.*

Tautologies and Contradictions

A (compound) statement is called a **tautology** if it is always true (regardless of the values of its component statements).

A (compound) statement is called a **contradiction** if it is always false (regardless of the values of its component statements).

Example:

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

$p \vee \sim p$ is a tautology

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

$p \wedge \sim p$ is a contradiction

Normally, **t** is used to denote a tautology and **c** is used to denote a contradiction.

Order

$\sim$
 $\downarrow$
 $\neg$ and $\vee$

$\downarrow$
 $\rightarrow$ and $\leftrightarrow$

Of course these order can be broken by using the bracket.

$$\text{Eg : } p \rightarrow q \vee \neg r \equiv p \rightarrow [q \vee (\neg r)]$$

$$(\underline{p \rightarrow q}) \vee \underline{\neg r} \equiv [\underline{p \rightarrow q}] \vee (\neg r)$$

Tautologies and Contradictions

Example:

$$(p \wedge q) \rightarrow (p \vee q)$$

hypothesis ↓

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

↓

$(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Tautologies and Contradictions

Example: $(p \wedge q) \wedge (\sim p \wedge r)$ is a contradiction.

P	q	r	$p \wedge q$	$\sim p$	$\sim p \wedge r$	$(p \wedge q) \wedge (\sim p \wedge r)$
T	T	T	T	F	F	F
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	T	F
F	T	F	F	T	F	F
F	F	T	F	T	F	F
F	F	F	F	T	F	F